Opgave: Dag 7 øvelse 1
(ELE1)

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$I_a := 8\ A$ $\Phi_p := 0.02\ Wb$ $p_p := 4$ $N_z := 800$ $M_v := 2$

Exercise 1

DC motor has the following specifications:

Armature current I_a = 8 A
Total flux per pole **Φ** = 0.02 Wb
Number of poles **P** = 4
Number of conductors in the armature winding **Z** = 800
Number of parallel paths in the winding **A** = 2

alculate the armature torque **Ta** generated by the otor

Del 1 - Maskinkonstanten

p er antal poler, N antal ledere og M antal parallelle veje.
Maskinkonstanten:

$$k_a = \frac{pN}{2\pi M} \quad (13.10)$$

$$k_T := \frac{p_p \cdot N_z}{(2 \cdot \pi \cdot M_v)} = 254.648$$

Del 2 - Ankermomentet

Momentet er proportionalt med flux og ankerstrøm:

$T = k_T \phi I_a$ for $\gamma = 90°$ DC machine torque (13.6)

$$T_a := k_T \cdot \Phi_p \cdot I_a = 40.744\ N \cdot m$$

Slidens form: $T_a = P \cdot \Phi \cdot Z \cdot I_a / (2\pi \cdot A)$ giver det samme:

$$\frac{p_p \cdot \Phi_p \cdot N_z \cdot I_a}{(2 \cdot \pi \cdot M_v)} = 40.744\ N \cdot m$$

$$T_a = 40.744\ N \cdot m$$

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